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G H Raisoni International Skill Tech University, Pune

S.Y B. Tech (TERM-IV)

CAE I- Summer 2026 (2024 Pattern)

Operating Systems (25BTPCL403)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) Attempt any one from Q1-Q2 and any one from Q3-Q4. ii) Neat diagrams must be drawn wherever necessary. iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed v) Assume suitable data, if necessary.

Q.No	Question	Marks	CO	BL
Q1.(a)	What are system calls? Explain the various categories of system calls.	[5]	I	2
b)	Determine types of Operating System	[5]	I	2
	OR			
Q 2.(a)	Summarize Evolution of Operating System	[5]	I	2
(b)	Explain Functions of Operating System	[5]	I	2
Q 3.(a)	Illustrate Process Scheduling Queues	[5]	II	3
(b)	What is a process? Draw and explain process state diagram	[5]	II	2
	OR			
Q. 4 (a)	Consider the following set of processes with their **Arrival Time (AT)** and **Burst Time (BT)**. Using the **First Come First Served (FCFS)** scheduling algorithm, calculate: 1. Gantt Chart 2. Turnaround Time (TAT) 3. Average Turnaround Time	[5]	II	3

Process	Arrival Time (AT)	Execute Time (BT)	Service Time (CT)
P0	0	5	0
P1	1	3	5
P2	2	8	8
P3	3	6	16

(b) What do you mean by PCB? What are its contents? Explain [5] II 2